SEQUOIA SECURITY

Drone-Enhanced Wildfire Monitoring & Fire Watch Services

Tagline: Beyond Traditional Fire Watch. Intelligence From Above.

Service Overview

Sequoia Security is developing a next-generation wildfire monitoring and fire watch service designed for utility companies, hydroelectric facilities, energy infrastructure, forestry operations, and remote industrial sites throughout British Columbia.

Traditional fire watch services rely primarily on security personnel conducting ground-level patrols, visually identifying smoke or fire, and reporting potential hazards.

Our approach takes fire watch operations beyond ground-level observation.

Under our proposed deployment model, every designated fire watch security guard will also be trained and certified as a Transport Canada Advanced Operations drone pilot and equipped with a DJI Mavic 3 Thermal (M3T) enterprise drone.

Combining ground patrols with aerial thermal imaging, our personnel will be able to inspect designated monitoring areas, identify potential heat anomalies, document fire-related hazards, and provide geographically referenced information to incident coordinators.

The objective is to transform conventional fire watch services into a technology-enabled wildfire intelligence operation.

Our Core Services

1. Advanced Fire Watch & Ground Security

Trained security personnel conduct scheduled patrols, observe potential ignition sources, monitor high-risk areas, and document site conditions. Personnel follow established reporting and emergency escalation procedures.

2. Thermal Drone Surveillance

Using DJI Mavic 3 Thermal drones, authorized operators conduct aerial monitoring of designated areas. Thermal imaging supports identification of potential hotspots, abnormal heat signatures, and areas requiring closer inspection.

3. Interactive Wildfire Mapping

Captured drone imagery and georeferenced field observations are integrated into a digital mapping platform.

The platform is intended to visualize:

- Potential fire hotspots and detected thermal anomalies.
- Observed or estimated fire-affected areas.
- Locations of critical infrastructure and assets.
- Security patrol and drone observation locations.
- Changes in observed fire conditions over time.
- Areas requiring additional monitoring or immediate escalation.

Maps are updated as new verified observations and processed imagery become available, providing near-real-time situational awareness where operational connectivity permits.

4. Live Aerial Situational Awareness

Where authorized and technically supported, drone operators can transmit live camera footage to designated monitoring personnel.

This provides an additional aerial perspective to assist with field coordination, risk assessment, and operational decision-making.

5. Emergency Documentation & Reporting

Each monitoring operation can generate structured reports containing flight records, thermal observations, geotagged imagery, incident locations, timestamps, and follow-up recommendations.

These records support communication, incident review, risk assessment, and seasonal wildfire monitoring documentation.

How Our Service Works

Step 1 — Ground Monitoring: Security personnel conduct routine patrols and assess potential fire hazards across designated facilities and surrounding areas.

Step 2 — Aerial Inspection: Where legally permitted and operationally safe, certified drone pilots deploy thermal-equipped aircraft to inspect areas of concern.

Step 3 — Thermal Detection: Drone cameras capture visible-light and thermal imagery to identify potential heat signatures and document observable fire-related conditions.

Step 4 — Digital Mapping: Observations are geotagged and incorporated into a centralized geographic monitoring dashboard.

Step 5 — Reporting & Coordination: Verified information is shared with authorized client representatives and relevant emergency authorities through established procedures.

Designed for Critical Infrastructure

Our proposed service is particularly suited for organizations operating infrastructure in remote, forested, or wildfire-prone environments.

Potential applications include:

- Hydroelectric dams, generating stations, and associated infrastructure.
- Transmission corridors and electrical substations.
- Remote utility facilities and construction projects.
- Industrial operations near forested areas.
- Forestry assets and other high-risk properties.

Why Sequoia Security?

Traditional fire watch observes. Sequoia Security is designed to observe, detect, map, and report.

By combining physical security, aerial thermal inspection, and geospatial intelligence, our proposed service aims to deliver:

- Expanded visibility of designated monitoring areas.
- Faster investigation of potential fire hazards.
- Better documentation of thermal anomalies.
- Location-specific mapping and incident reporting.
- Enhanced situational awareness for infrastructure operators.
- One integrated field team instead of separately contracted security and drone monitoring personnel.

Our Commitment

Sequoia Security aims to provide utility and industrial clients with a practical, scalable wildfire monitoring solution that combines human judgment, professional security operations, advanced aerial technology, and geospatial data.

All drone operations will be subject to Transport Canada requirements, applicable airspace restrictions, site permissions, and coordination with the appropriate wildfire authorities.

Sequoia Security — Protecting Critical Infrastructure From the Ground and the Sky.

Call to Action: Request a Wildfire Monitoring Proposal